Team 19 – Tigers

Competition: ICDC 2026
Team Role: Blue Team

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1. Executive Summary

This document outlines the architecture, configuration, and security posture of Team 19's competition network. The key elements covered are:

- Network topology and addressing
- Server roles and service architecture
- Active Directory (AD) configuration
- Web service restoration
- Firewall and port controls
- Application deployment (Tickets API)
- Security hardening measures
- Operational readiness

Objectives:

- Restore required services
- Secure infrastructure against attacks
- Ensure service persistence and stability
- Reduce attack surface while maintaining availability

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2. Network Architecture

2.1 Logical Layout

The network consists of the following segmented service hosts, with their corresponding **IPv6** and **IPv4** addresses:

Host	FQDN	Primary Role	IPv6 Address	IPv4 Address	Ports Required
AD	ad.team19.isucdc.com	Domain Controller / CA	`fe80::a85c:e705:597d:67b3%5`	`197.45.10.10`	389, 636, 3389
TICKETS	tickets.team19.isucdc.com	Ticketing API (.NET Core)	`fe80::a1a7:bf2a:51b8:acd3%6`	`197.45.10.20`	5000, 3389
MGMT	mgmt.team19.isucdc.com	Management Node	`fe80::be24:11ff:fed8:e599`	`197.45.10.30`	5000, 22
ADMIN	admin.team19.isucdc.com	Admin Web Interface (IIS)	`fe80::d5f8:c607:730e:f304%8`	`197.45.10.40`	80, 3389
WWW	www.team19.isucdc.com	Public Web Server	`fe80::be24:11ff:fe2e:d6c0`	`197.45.10.50`	80, 22

2.2 Network Design Decisions

- **Internal DNS** provided by AD
- **Fully qualified domain names** enforced
- **All services** bound to internal IP space
- **Firewall rules** strictly limited to required ports
- **No unnecessary service exposure**

2.3 Diagram (Logical Representation)

```
              [ AD - Domain Controller ]
              389 / 636 / 3389
                    |
  -------------------------------------------------------
  |          |          |          |
[TICKETS]   [ADMIN]    [WWW]      [MGMT]
5000,3389   80,3389    80,22      5000,22
```

All services rely on AD for:

- DNS resolution
- Authentication
- Certificate services (LDAPS)

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3. Active Directory Configuration

3.1 Domain

- **Domain:** team19.isucdc.com
- **Services verified:**
  - LDAP (389)
  - LDAPS (636)
  - RDP (3389)
  - DNS
- **Verification Tools Used:**
  - dcdiag
  - nltest /dsgetdc
  - certutil -config - -ping

All core AD diagnostics passed.

3.2 Administrative Structure

- **New Administrative Account:**
  - Username: ad-admin-01
  - Group: Domain Admins
  - Purpose:
    - Avoid use of built-in Administrator
    - Enable audit tracking of privileged activity
    - Improve accountability

3.3 AD Hardening Measures

1. **LDAP Integrity Enforcement**: Ensured proper LDAP signing configuration.
2. **SMBv1 Disabled**: Prevents downgrade and exploit risks.
3. **Print Spooler Disabled (DC)**: Mitigates PrintNightmare exploitation risk.

4. **Remote Registry Disabled**: Reduces lateral movement opportunities.
5. **Password Policy Hardened (via GPO)**:
    - Minimum Length: 14
    - Complexity: Enabled
    - Lockout Threshold: 5 attempts
    - Lockout Duration: 15 minutes

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4. Tickets Service (Port 5000)

4.1 Application Type

- **Platform**: ASP.NET Core Web API
- **Runtime**: .NET
- **Server**: Kestrel
- **Directory**: C:\Users\Administrator\Desktop\tickets
    - Contains: tickets.csproj, Program.cs, Controllers, appsettings.json

4.2 Restoration Steps

- **Issue**: Tickets API was not running initially.
- **Steps Performed**:
    - Verified no service listening on port 5000.
    - Built project: dotnet build
    - Ran application: dotnet run --urls=http://0.0.0.0:5000
    - Verified health endpoint: curl http://localhost:5000/api/tickets/health (Result: HTTP 200 OK)
    - Confirmed service listening on 0.0.0.0:5000.

4.3 Firewall Configuration

- **Inbound rule added:**
  New-NetFirewallRule -DisplayName "Allow TCP 5000" -Direction Inbound -Protocol TCP -LocalPort 5000 -Action Allow

Ensures access for the scoring engine and internal hosts.

4.4 Service Validation

- **Validated using:**
  netstat -an | findstr :5000
- **Result:**
  0.0.0.0:5000 LISTENING

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5. Admin Web Server (Port 80)

- **Platform**: IIS (World Wide Web Publishing Service)
- **Installation Commands:**
  - Install-WindowsFeature Web-Server -IncludeManagementTools
  - Start-Service W3SVC

Firewall Configuration

- **Inbound rule added:**
  New-NetFirewallRule -DisplayName "Allow HTTP 80" -Direction Inbound -Protocol TCP -LocalPort 80 -Action Allow
- **Service Tested:**
  curl http://localhost

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6. Web Services (WWW)

- **Operational Services:**
  - HTTP (80)
  - SSH (22)
- **Security Measures:**
  - No unnecessary services exposed.

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7. Firewall Strategy

Principle: Least Privilege Networking

Allowed Ports:

- **LDAP**: 389
- **LDAPS**: 636
- **RDP**: 3389

- **HTTP**: 80
- **Tickets API**: 5000
- **SSH**: 22

All other inbound traffic is implicitly denied.

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8. Security Controls Implemented

8.1 Service Minimization

- Disabled unnecessary Windows services.

8.2 Role Separation

- AD, Tickets, Admin, WWW, and MGMT separated into dedicated hosts.

8.3 TLS Security

- **LDAPS verified operational** using internal CA.

8.4 Domain Authentication Enforcement

- All internal service authentication tied to AD.

8.5 Privileged Account Separation

- Created a new admin account and avoided default Administrator usage.

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9. Intrusion Reduction Strategy

- **No IDS appliance provided**, but the following were implemented:
    - Strict firewall port limitation
    - Removal of legacy protocols
    - Reduced lateral movement vectors
    - Service binding control (0.0.0.0 vs localhost)
    - Account lockout policy

Future Planned Additions (if time allowed):

- Sysmon deployment
- Centralized event log forwarding
- Windows Defender ASR rules

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10. Operational Readiness

All critical services verified:

Service	Status
AD LDAP	Operational
AD LDAPS	Operational
AD RDP	Operational
Tickets API	Operational
Admin HTTP	Operational
WWW HTTP	Operational

Remaining web authentication checks are under active configuration.

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11. Risk Assessment

Primary Risks Identified:

- Application-layer vulnerabilities
- Credential-based attacks
- Lateral movement attempts
- RDP brute force attempts

Mitigation Strategies:

- Strong password enforcement
- Limited exposed ports
- Service isolation
- Reduced DC attack surface

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12. Conclusion

Team 19 has:

- Restored all core infrastructure services
- Hardened Active Directory
- Deployed and validated Tickets API

- Secured web services via firewall policy
- Implemented structured administrative controls
- Established a defensible and auditable architecture

Our environment balances availability and security and is prepared for the active attack phase.